

Master of Computer Application – Game Development
Semester – I

Sub: Front-End Frameworks and Technologies

Topic: Weather Information App

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INTRODUCTION

The Weather Information App is a client-side web application developed using HTML5, CSS3, and JavaScript to fetch and display real-time weather details for any city in the world. It allows users to enter a city name and instantly view the current temperature, humidity, wind speed, and weather condition, along with a matching weather icon.

The application uses JavaScript functions, conditional statements, DOM manipulation, event handling, and the Fetch API with Promises to retrieve live data from the OpenWeatherMap API in JSON format. It also includes input validation, a loading indicator, and error handling for invalid or unknown city names, along with a responsive CSS Grid layout that adapts to both desktop and mobile screens.

Overall, this project demonstrates the practical use of API integration, asynchronous programming, dynamic content rendering, and responsive web design in creating a simple and user-friendly weather information system.

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INPUT CODE

File-Weather.htm

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Weather Information App</title>

    <link rel="stylesheet" href="Weather.css">
</head>

<body>

    <div class="container">

        <h1>Weather Information</h1>

        <input
            type="text"
            id="city"
            placeholder="Enter city name"
        >

        <button onclick="getWeather()">Search</button>

        <p id="loading"></p>

        <div id="weather">

            <h2 id="cityName"></h2>
```

```
            <img id="weatherIcon" src="" alt="Weather Icon">
            <p id="temperature"></p>
            <p id="humidity"></p>
            <p id="wind"></p>
            <p id="condition"></p>
        </div>
        <p id="error"></p>
    </div>
    <script src="Weather.js"></script>
</body>
</html>
```

File-Weather.css

```
* {
    box-sizing: border-box;
}

body {
    margin: 0;
    padding: 45px 20px;
    min-height: 100vh;
    color: white;
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #08152d, #15375f);
}
```

```
.container {
    width: 90%;
    max-width: 900px;
    min-height: 600px;
    margin: auto;
    padding: 50px;
    border: 1px solid #3c628b;
    border-radius: 25px;
    background: #102d55;
    box-shadow: 0 15px 35px rgba(0, 0, 0, 0.35);
}

h1 {
    margin: 0 0 30px;
    font-size: 48px;
    font-weight: normal;
}

#city {
    width: calc(100% - 130px);
    height: 52px;
    padding: 0 16px;
    color: white;
    font-size: 16px;
    border: 1px solid #507da8;
    border-radius: 10px;
    outline: none;
    background: #0a2142;
}

#city::placeholder {
    color: #91a7c3;
}

#city:focus {
    border-color: #6cc9ff;
```

```
}

button {
    width: 115px;
    height: 52px;
    margin-left: 10px;
    color: #08203c;
    font-size: 16px;
    font-weight: bold;
    border: none;
    border-radius: 10px;
    cursor: pointer;
    background: #65c4f7;
}

button:hover {
    background: #88d5ff;
}

#loading,
#error {
    min-height: 20px;
    margin: 15px 0 0;
    font-size: 14px;
}

#loading {
    color: #8bd4ff;
}

#error {
    color: #ffb5bc;
}

/* Weather information card */
#weather {
    display: grid;
```

```
    grid-template-columns: 1fr 1fr;
    gap: 18px 25px;
    margin-top: 30px;
    padding: 30px;
    border: 1px solid #4779a5;
    border-radius: 20px;
    background: #1b507e;
}

#cityName {
    grid-column: 1 / 3;
    margin: 0;
    font-size: 32px;
}

#weatherIcon {
    width: 125px;
    height: 125px;
    object-fit: contain;
}

#temperature {
    align-self: center;
    margin: 0;
    font-size: 48px;
    font-weight: bold;
}

#humidity,
#wind,
#condition {
    margin: 0;
    padding: 16px;
    color: #b7c9dd;
    font-size: 15px;
    line-height: 1.5;
    border: 1px solid #3d6c96;
```

```
    border-radius: 10px;
    background: #12365f;
}

#humidity,
#wind {
    grid-column: span 1;
}

#condition {
    grid-column: 1 / 3;
}

@media (max-width: 600px) {
    body {
        padding: 20px 10px;
    }

    .container {
        width: 100%;
        min-height: 100vh;
        padding: 30px 20px;
    }

    h1 {
        font-size: 36px;
    }

    #city {
        width: 100%;
    }

    button {
        width: 100%;
        margin: 10px 0 0;
    }
```

```
    #weather {
        padding: 20px;
    }

    #cityName,
    #condition {
        grid-column: 1 / 3;
    }

    #weatherIcon {
        width: 90px;
        height: 90px;
    }

    #temperature {
        font-size: 34px;
    }
}
```

File-Weather.js

```
const API_KEY = "f914f86f20870ab64bac31c246db975b";

function getWeather() {

    let city = document.getElementById("city").value;

    if (city === "") {

        document.getElementById("error").innerHTML =
            "Please enter a city name.";

        return;
    }

    document.getElementById("loading").innerHTML =
        "Loading...";
```

```
    document.getElementById("error").innerHTML = "";

    let url =
        `https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${API_KEY}&units=metric`;

    fetch(url)

        .then(function(response) {

            if (!response.ok) {
                throw new Error("City not found");
            }

            return response.json();
        })

        .then(function(data) {

            console.log(data);

            document.getElementById("cityName").innerHTML =
                data.name;

            document.getElementById("temperature").innerHTML =
                "Temperature: " + data.main.temp + " °C";

            document.getElementById("humidity").innerHTML =
                "Humidity: " + data.main.humidity + "%";

            document.getElementById("wind").innerHTML =
                "Wind Speed: " + data.wind.speed + " m/s";

            document.getElementById("condition").innerHTML =
                "Condition: " + data.weather[0].description;
```

```
            let icon = data.weather[0].icon;

            document.getElementById("weatherIcon").src =
                `https://openweathermap.org/img/wn/${icon}@2x.png`;

            document.getElementById("loading").innerHTML = "";
        })

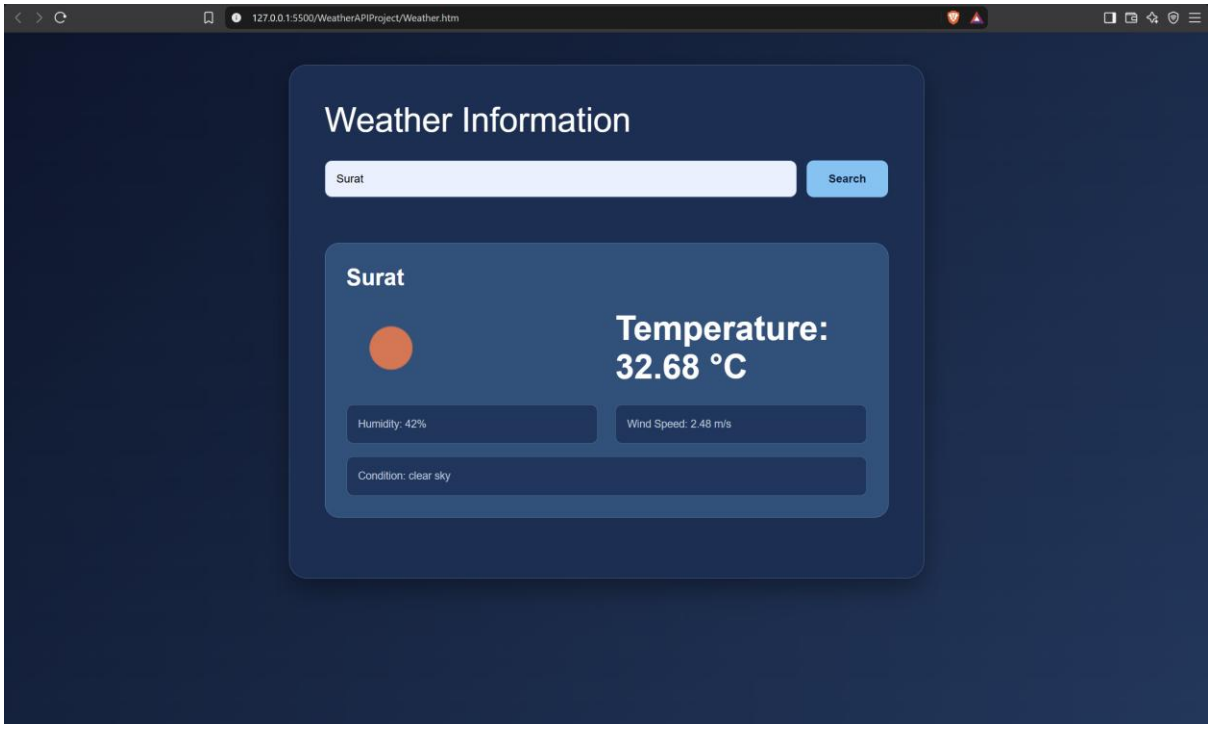
        .catch(function(error) {

            document.getElementById("loading").innerHTML = "";

            document.getElementById("error").innerHTML =
                "City not found. Please try again.";

            console.log(error);
        });
}
```

OUTPUT


127.0.0.1:5500/WeatherAPIProject/Weather.htm
Weather Information
Surat
Search
Surat
Temperature:
32.68 °C
Humidity: 42%
Wind Speed: 2.48 m/s
Condition: clear sky

CONCLUSION

The Weather Information App successfully demonstrates how HTML, CSS, and JavaScript can be combined to create a simple, interactive, and user-friendly weather lookup system. By using JavaScript to read the city name, send a request to the OpenWeatherMap API, and display the response, the application provides an efficient way to view live weather details without requiring any server-side processing of its own.

The system effectively retrieves and processes data using the Fetch API, Promises, and JSON objects, while features such as input validation, a loading indicator, error handling for invalid city names, and weather icons improve usability and interaction. The use of CSS Grid and media queries also allows the interface to adapt smoothly to both desktop and mobile screens.

Overall, this project demonstrates fundamental web development concepts such as API integration, asynchronous programming, DOM manipulation, event handling, functions, conditional statements, dynamic content rendering, and responsive design in creating a practical weather information application.